

Exact Closed-Form Least-Squares Projection for State Space Model Parameter Updates

Replacing Iterative Gradient Steps with Algebraic Solutions
via Equality-Constrained Quadratic Programming

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Problem Statement and Literature Context

State Space Models (SSMs), and in particular MAMBA [2], represent the current state of the art in sequence modeling as efficient alternatives to Transformer architectures [3], achieving linear-time complexity in sequence length while matching or exceeding Transformer performance on a range of language modeling benchmarks [2, 4]. Training these models relies uniformly on first-order iterative optimizers — most prominently Adam [7] — applied to all parameters simultaneously, including components whose mathematical structure does not require iteration.

The present proposal identifies one such component. The output projection matrix C in every MAMBA block satisfies, given a fixed batch of hidden states X and targets Y , a least-squares optimality condition with a positive semidefinite Hessian. This is precisely the class of problems analyzed in Lin & Liang [1], which admit a derivative-free, closed-form, exact algebraic solution. The practical consequence is that Adam — and by extension any iterative first-order or quasi-Newton method applied to this subproblem [8] — introduces approximation error at every step and spends iterations converging to something that is algebraically available in one step.

This gap between the algebraic structure of the problem and the numerical method applied to it has not been identified in the SSM literature. Existing training pipelines for MAMBA [2], S4 [4], and related state space architectures [5, 6] apply Adam uniformly to all parameters, treating the linear output projection identically to the nonlinear selective scan mechanism. This is sub-optimal: the closed-form solution framework of Lin & Liang [1] provides not only an exact solution in one step, but a complete characterization of the solution set — information that iterative methods cannot produce.

The relevance to the literature is twofold. First, hybrid training protocols that combine exact solvers for linear subproblems with iterative methods for nonlinear components have a precedent in classical numerical optimization [8] and in the neural network literature on Gauss–Newton and natural gradient methods [9, 10]. Second, the algebraic characterization of the solution set — particularly the null-space structure — provides diagnostic information about training dynamics that has no analogue in gradient-based analysis [1]. The proposed research connects these two lines of work by applying the exact QP framework of Lin & Liang [1] to a state-of-the-art sequential architecture for the first time.

Preliminary: Unconstrained Formulation

Before introducing the equality constraint it is useful to establish the unconstrained baseline. Consider the output projection of a MAMBA layer:

$$\hat{Y} = XC^\top, \quad X \in \mathbb{R}^{T \times n}, \quad C \in \mathbb{R}^{d \times n}, \quad Y \in \mathbb{R}^{T \times d}.$$

For each column $c_j \in \mathbb{R}^n$ of $C^\top$, the optimal update minimizing $\|Xc_j - y_j\|_2^2$ satisfies the normal equations $(X^\top X)c_j^* = X^\top y_j$. Setting $P = X^\top X \succeq 0$, $q = -X^\top y_j$, and $s = 0$, this is the unconstrained convex quadratic function

$$F_x(c_j) = \frac{1}{2} c_j^\top P c_j + q^\top c_j$$

in the notation of Lin & Liang (2023), equation (2.1), with $x \equiv c_j$. The unconstrained closed-form solver provides the necessary and sufficient condition for a finite optimum — $q \in \mathcal{R}(P)$, i.e., $X^\top y_j \in \mathcal{R}(X^\top X)$ — and the minimum-norm solution:

$$c_j^* = -(X^\top X)^\dagger (X^\top y_j) + \tilde{\varepsilon}, \quad \tilde{\varepsilon} \in \mathcal{N}(X^\top X),$$

where $\tilde{\varepsilon} = 0$ selects the minimum-Frobenius-norm solution. This result establishes the algebraic foundation; it is a prerequisite, not the main contribution.

Main Contribution: Equality-Constrained Formulation

Anchor Constraints

The central contribution is the introduction of *anchor constraints* within the exact QP framework. Let $\mathcal{A} \subset \{1, \dots, T\}$ be a set of k token positions from the current training batch. The physical motivation is concrete: these are positions at which the model's prediction is already reliable and must be preserved exactly after the weight update.

Defining $X_{\mathcal{A}} \in \mathbb{R}^{k \times n}$ and $Y_{\mathcal{A}} \in \mathbb{R}^{k \times d}$ as the rows of X and Y at the anchor positions, the equality constraint requires $CX_{\mathcal{A}}^\top = Y_{\mathcal{A}}^\top$. For each output dimension j the update problem becomes:

$$\min_{c_j \in \mathbb{R}^n} \|Xc_j - y_j\|_2^2 \quad \text{subject to} \quad X_{\mathcal{A}} c_j = y_{\mathcal{A},j},$$

which is exactly Problem (3.9) of Lin & Liang [1], with:

$$P = 2X^\top X \succeq 0, \quad q = -2X^\top y_j, \quad s = \|y_j\|^2, \quad A = X_{\mathcal{A}}, \quad b = y_{\mathcal{A},j}.$$

Closed-Form Solution

Following Section 3.2 of Lin & Liang [1], let the SVD of $A = X_{\mathcal{A}}$ yield $V_2 \in \mathbb{R}^{n \times (n-k)}$ spanning $\mathcal{N}(X_{\mathcal{A}})$. The closed-form particular solution is:

$$c_j^* = A^\dagger b - V_2 (V_2^\top P V_2)^\dagger V_2^\top (q + P A^\dagger b) + V_2 \varepsilon^*, \quad \varepsilon^* \in \mathcal{N}(V_2^\top P V_2),$$

with $\varepsilon^* = 0$ selecting the minimum-norm solution within the feasible affine subspace. The optimum is unique if and only if $\mathcal{N}(A) \cap \mathcal{N}(P) = \{0\}$.

Analytical Insights

The closed-form solution provides qualitatively different information from what any iterative solver can produce.

Uniqueness characterization. Classical numerical solvers [8] and Adam [7] terminate at a single approximate point with no information about whether it is the unique minimizer or one element of a continuous family of equally optimal weight matrices. The closed-form solver answers this exactly: the solution is unique if and only if $\mathcal{N}(X_{\mathcal{A}}) \cap \mathcal{N}(X^\top X) = \{0\}$, verifiable in $\mathcal{O}(Tn)$ via SVD.

Verifiable optimality condition. The existence of a finite optimum under the equality constraint is governed by a necessary and sufficient algebraic condition on the data matrices, computable before any optimization is performed [1]. Neither Adam nor classical quasi-Newton solvers [8] have an equivalent.

Characterization of invisible directions. The columns of V_2 beyond the rank of X_A span directions in weight space where both the gradient of the objective and the constraint residual are identically zero. These directions are completely invisible to gradient-based methods [9]. The exact solver identifies and parametrizes them completely, providing an algebraic certificate of optimality unavailable from any iterative approach.

Before and after. Before: iterative solvers produce one approximate numerical point, with no information about uniqueness, solution freedom, or the structure of the optimal set. After: the complete set of optimal solutions is characterized in closed form, a certificate of optimality is available, and all remaining degrees of freedom are explicitly identified. The output of any iterative method is one specific element of this set; which one, and why, is a question only the closed-form solution can frame.

Main Claim

Main Claim 1. *The output projection update in a MAMBA layer, when computed via the equality-constrained closed-form QP solver of Lin & Liang [1] applied to $\min_C \|XC^\top - Y\|_F^2$ subject to $CX_A^\top = Y_A^\top$, achieves the constrained residual $\|XC^* - Y\|_F$ at machine precision in a single algebraic step. In contrast, classical iterative numerical solvers [8] — including Adam [7] — require a measurable number of iterations to achieve an equivalent residual: empirically, more than 50 steps are needed to approach 10^{-4} , and 10^{-8} is not reached within 500 steps for any tested learning rate. Furthermore, incorporating periodic exact updates into the training loop will produce a measurably faster decrease in perplexity on WikiText-2 compared to the Adam baseline; and the number of anchor positions k is identifiable as a parameter controlling the trade-off between solution uniqueness and residual magnitude.*

Algebraic Mechanisms

Singular Value Decomposition of X and X_A

Two SVD computations are central. The SVD of X provides $(X^\top X)^\dagger$ and characterizes the null-space freedom of the unconstrained problem. The SVD of X_A provides $A^\dagger$ and the null-space basis V_2 , which determines the feasible affine subspace and the residual solution freedom after the anchor constraint is imposed.

Null Space and Solution Freedom

When $T > n$ (overdetermined batch), $\mathcal{N}(X^\top X) = \{0\}$ and the solution is always unique regardless of k — confirmed empirically with $T = 2048$, $n = 512$, $\kappa(X^\top X) = 69.1$. The underdetermined regime ($T < n$), where the null space is nontrivial, is the algebraically richer case and is accessible by reducing the batch size below n .

Moore–Penrose Pseudoinverse

The pseudoinverse enters twice: as $(X^\top X)^\dagger$ in the unconstrained baseline and as $(V_2^\top P V_2)^\dagger$ in the equality-constrained solver. In both cases it selects the minimum-Frobenius-norm element from the solution set without any hyperparameter, as established in Lin & Liang [1].

Positive Semidefiniteness of $P = 2X^\top X$

By construction $P \succeq 0$ always. The reduced Hessian $V_2^\top P V_2 \succeq 0$ inherits positive semidefiniteness; its rank determines the dimension of the residual solution freedom. The condition number $\kappa(X^\top X)$ is monitored throughout training in Experiment 5.

Methodology and Implementation

Isolating the Target Component

The starting point is `mamba-minimal` [11], a pure PyTorch reimplementation of MAMBA in approximately 300 lines, trained on WikiText-2 [12]. The output projection layer of block 1 is isolated via a forward hook on `out_proj`. All matrices are cast to `float64` before algebraic computations. Rank and feasibility are verified via SVD before any optimization.

Experiment 1 — Residual Comparison

The exact solver is compared against Adam with learning rates in $\{10^{-2}, 10^{-3}, 10^{-4}\}$ for 500 iterations initialized at $C = 0$. Empirical result: the exact solver reaches $\|XC^* - Y\|_F = 2.70 \times 10^{-14}$ in one step; Adam ($\text{lr} = 10^{-3}$) reaches 1.75×10^{-4} after 500 steps and does not reach 10^{-8} within the allotted iterations. The comparison with classical numerical methods (conjugate gradient, L-BFGS [8]) is included to contextualize the result beyond the Adam baseline. Output: `exp1_residual_comparison.png`.

Experiment 2 — Anchor Sweep

k is varied over $\{1, \lfloor n/4 \rfloor, \lfloor n/2 \rfloor, n-1\}$ for three anchor selection strategies: random, low-uncertainty, and high-uncertainty. Residual, constraint violation, solution freedom dimension, and uniqueness flag are recorded at each $(k, \text{strategy})$ combination. Output: `exp2_anchor_sweep.png`.

Experiment 3 — Null Space Verification

The singular value spectrum of $X^\top X$ is computed and the algebraic certificate $\max_j |g_j^\top v| < 10^{-6}$ is verified for all null-space vectors. Output: `exp3_sv_spectrum.png`.

Experiment 4 — Hybrid Training Protocol

The exact solver is integrated as a periodic correction every $k_{\text{step}} \in \{5, 10, 20, 50\}$ Adam steps. Training runs for 1000 steps on WikiText-2; perplexity is recorded every 50 steps. Output: `exp4_perplexity.png`.

Experiment 5 — Algebraic Monitoring

Rank, condition number, null-space dimension, and uniqueness flag of $X^\top X$ are recorded at each exact-correction step throughout training. Output: `exp5_monitoring.png`.

Expected Results

Primary result. The exact solver achieves $\|XC^* - Y\|_F \approx \varepsilon_{\text{machine}}$ in one step (2.70×10^{-14} empirically). Classical iterative solvers, including Adam [7] and L-BFGS [8], require more than 50 steps to approach 10^{-4} and do not reach 10^{-8} within 500 steps — a gap of nine orders of magnitude relative to the exact solver.

Structural result. Variation of k identifies the exact transition between non-unique and unique solutions, determined algebraically rather than estimated from a convergence criterion.

Training result. The hybrid protocol reaches a target perplexity threshold in fewer steps than pure Adam, with discrete improvements at each exact-correction event.

Algebraic result. Singular value monitoring reveals which batches produce a singular Gram matrix and whether this correlates with training instability — invisible to gradient-based diagnostics.

All results are fully reproducible: fixed random seeds, WikiText-2, NumPy and SciPy as minimum dependencies.

Relevance to the Professor’s Research Agenda

Prof. Lin’s central contribution in Lin & Liang [1] is demonstrating that problems conventionally solved by iterative numerical methods admit exact algebraic solutions when their mathematical structure is properly recognized. The present proposal applies this philosophy to sequential neural architectures — a domain where the structure has not previously been exploited, despite being present and directly applicable.

The connection is not analogical. The same equality-constrained closed-form solver, applied to the algebraic object ($P = X^\top X \succeq 0$, $q = -X^\top y_j$, $A = X_{\mathcal{A}}$, $b = y_{\mathcal{A},j}$), produces the same closed-form parametrization of all optimal solutions via $\mathcal{R}(P)$, $\mathcal{N}(P)$, and $\mathcal{N}(A)$. This extends the domain of applicability of Lin & Liang [1] from classical quadratic programming to state-of-the-art neural architectures, which is precisely the direction motivated in the paper’s conclusion.

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